Project Initialization and Planning Phase

Date	15 July 2026
Team ID	SWTID-2026-2089
Project Title	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

This project develops a machine learning solution to predict whether a credit card application should be approved or rejected. It improves decision-making by analyzing applicant information and credit history, reducing manual effort and improving consistency.

Section	Details
Objective	Develop a Random Forest based model to predict credit card approval accurately and efficien
Scope	Preprocess application and credit data, train the ML model, evaluate performance, and suppo
Problem Description	Manual credit approval is time-consuming, inconsistent, and prone to human error.
Impact	Improves approval speed, reduces operational cost, supports fair decision-making, and enhar
Approach	Use data preprocessing, feature engineering, Random Forest classification, and model evalua
Key Features	• ML-based approval prediction • Data preprocessing • Feature importance • Confusion Matrix & ROC Curve • Trained model export (.pkl)

Resource Requirements

Resource	Description	Specification
Hardware	CPU/RAM	Intel i5 or higher, 8GB RAM
Storage	Disk Space	1 GB+
Framework	Backend	Python, Flask
Libraries	ML	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development	Environment	Google Colab / Jupyter Notebook
Dataset	Source	application_record.csv & credit_record.csv